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Artificial Intelligence

Detection of Non-Authentic Scientific Papers

SCIgen Case and More Advanced Generating Methods

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1 Introduction

In recent decades, scientific output has grown rapidly under the “publish or perish” dynamic, which frequently puts quantity over quality. This aphorism refers to the pressure academics have to publish their work in order to advance in their careers. Is an issue that has become more pronounced in the modern era due to the massive growth of digital platforms, and it has increased the risk of information saturation.

Furthermore, as artificial intelligence has advanced, it has become more frequent for papers to incorporate automated language, such as that produced by Chat-GPT, which is frequently challenging to detect within the text itself.

In 2005, there were already techniques available that could produce whole fake articles. In this project, the focus is on SCIGen; with the use of context-free grammar, computer science terminology, graphs, diagrams, and citations, the free software can produce nonsensical texts.

This study will examine different classification models aimed at identifying fake papers produced by SCIGen and also by more advanced systems, such as Large Language Models. These will be compared with authentic articles from arXiv.

The SCIGen software is no longer available, and most of the publications have been removed from the internet.

1.1 Methodology

The methodology adopted in this paper follows an experimental, supervised-learning approach aimed at developing and validating an artificial intelligence model capable of distinguishing authentic articles from automatically generated ones. To achieve this goal, it was necessary to integrate several steps, from the construction of the dataset to the design of the computational pipeline, up to the analysis of the results. In particular, the study combines normalization techniques with multiple machine learning classifiers, and the models are systematically compared through standard performance metrics such as accuracy, precision, recall, and F1-score. In addition, is also includes an n-gram analysis aimed at identifying recurring lexical patterns that differentiate authentic from generated articles.

2 Detecting SCIgen generated papers

The focus of this chapter is to identify fake papers that are only produced by SCIgen, a software designed by three doctoral students at the Massachusetts Institute of Technology (MIT) in 2005. The aim was to demonstrate that conferences would accept meaningless papers and that fun would be maximized at the expense of coherence (Ball, 2005).

The dataset, text pre-processing steps, and classification criteria must all be explained before putting into practice the classification model that can distinguish between automated and real articles.

2.1 Dataset

Gathering and preparing the dataset is part of the preliminary phase, which is essential for the project's success. The corpus is built around two main categories:

- Authentic articles from reliable scientific databases, such as arXiv;
- Fake articles obtained using SCIgen software.

It should be noted that the majority of the fictitious papers produced by SCIgen have already been deleted to preserve the accuracy of the data provided by publication services, such as Google Scholar. This will then influence the models given the limited range of options available for the research.

Therefore, the only documents available for this study are 100 and all "written" by Ike Antkare (a fictitious author).¹

In order to make the comparison methodologically more robust, the selection of real articles was done by prioritizing subject areas that were consistent with those that the software usually simulated, such as computer science and more specifically: cluster computing, operating systems, network and Internet architecture.

To prevent imbalances that can affect model training, special attention was paid to balancing the dataset: the proportion of produced and authentic publications was kept as close as possible.

The true scientific articles were downloaded from the computer science part of the arXiv website², specifically from the cs.NI, cs.DC, and cs.OS sections. Where:

¹https://lig-membres.imag.fr/labbe/Publi/IkeAntkare/Ike_AntKare_index.html

²<https://arxiv.org>

- cs.NI: Networking and Internet Architecture;
- cs.DC: Distributed, Parallel, and Cluster Computing;
- cs.OS: Operating Systems.

This subcategory selection guarantees a homogeneous sample where the subjects of the fictitious and authentic documents are consistent.

In total, 200 papers made up the dataset, half of which were fake and the other half were real. Each document was given a label so that classification models could be constructed: 0 if the paper is generated, and 1 if it is not.

2.2 Pre-processing

The collected texts, that had already been extracted, were next subjected to linguistic pre-processing, which was necessary to standardize the content and allow for efficient automated analysis. So, the raw data are converted into a structured format that the models can comprehend.

In this phase, a number of methods were used to filter out all the useless information or noise ones that might confuse the models. In particular the function `clean_text_spacy(text)` has been defined, which contains:

- Text Normalization: all text has been converted to lowercase and cleared of special characters, punctuation, and numerals (0–9).
- Tokenization: word tokenization was done using the spaCy package, which provides trustworthy linguistic segmentation.
- Stopwords: common English stopwords were eliminated by combining the NLTK and spaCy stopword collections.
- Lemmatization: to reduce inflected or derived versions of words to their basic lemma in order to limit the dimensionality of the feature space (e.g., running → run).

Listing 1: Text pre-processing

```

1 def clean_text_spacy(text):
2     """Clean the input text using spaCy for tokenization
3       and lemmatization."""
4
5     # Convert text to lowercase
6     text = text.lower()
```

```

7      # Remove digits
8      text = re.sub(r'\d+', '', text)
9
10     # Remove dashes between words
11     text = re.sub(r'(?<=\w)[- \u2010\u2011](?=\w)', ' ', text)
12
13     # Remove punctuation
14     text = re.sub(r'["'#$%&'*,.-/:;<=>?@\[\]\^_
15         \{|}~]"', '', text)
16
17     # Tokenize the text using spaCy
18     doc = nlp(text)
19
20     # Extract tokens, filtering out stopwords and non-
21     alphabetic tokens
22     tokens = [token for token in doc if token.is_alpha
23         and token.text not in stop_words]
24     tokens = [token for token in tokens if token not in
25         stop_words and len(token) > 1]
26
27     # Lemmatize the tokens
28     cleaned_text = lemmatize_text(tokens)
29
30     return cleaned_text

```

Finally, the clean tokens were recombined into processed documents, ready for vectorization (e.g., TF-IDF) and subsequent model training.

Furthermore, there was a clear pattern that could be used to identify the fake papers because they were all created by the same "author". This led to the creation of a list of unique stop words, which included "Ike," "Ike Antkare," and "Antkare." These combinations were then eliminated in order to prevent them from affecting the models.

The **Term Frequency – Inverse Document Frequency (TF - IDF)** representation was then used to vectorized the cleaned documents. In particular it balances:

- The local importance of a term within a document (TF).
- It's global rarity across the corpus (IDF).

The resulting sparse, high-dimensional vectors provide a suitable input representation for machine learning classifiers in text classification tasks.

2.3 Classification Models

As previously said, the project’s goal is to identify non-authentic scientific publications. To do this, several classification models were used and evaluated using multiple criteria. Following that, each method’s results were shown, and their respective performances were compared. The approaches chosen vary from the conventional ones to those with a more complex methodology. The models chosen for this study are the following:

1. **Support Vector Machine(SVM)**: applied due to its robustness in handling high-dimensional spaces and its effectiveness in text classification tasks.
2. **Logistic Regression**: chosen as a simple linear baseline model that is easy to interpret and performs well with high-dimensional sparse features such as TF-IDF.
3. **Naïve Bayes**: included as a standard baseline for text classification. Its probabilistic framework is particularly effective with word frequency features such as TF-IDF, despite its simplifying independence assumption.
4. **Random Forest**: selected as a representative ensemble method capable of capturing nonlinear relationships and reducing overfitting by averaging across multiple decision trees.

2.4 Results

As mentioned above, all of the classification models in the previous section utilized TF-IDF as input. To evaluate the performance of each, a confusion matrix was then generated. This is an helpful tool since it does not merely indicate total accuracy but demonstrates the nature of the model’s misclassifications (i.e., which observations are incorrectly classified). The performance of a model is done by comparing predicted labels with true labels:

- True positive (TP): positive classes correctly predicted.
- False positive (FP): negative classes incorrectly predicted as positive.

- True negative (TN): negative classes correctly predicted.
- False negative (FN): positive classes incorrectly predicted as negative.

Further performance indicators can be computed using the confusion matrix outcomes to provide a more comprehensive understanding of model behavior, such as precision, recall, and F1 score.

In Table 1 only the SVM model’s results are shown. However, all models yield the same conclusions in this case, as will be clear from the ROC curve visualizations.

Before examining the outcomes, it is important to first explain how the models were trained. The data set was split into two parts: 60% training data (120 observations, equally divided into 60 fake and 60 authentic papers) and 40% test data (80 observations, evenly divided into 40 fake and 40 real papers). Each model is trained using the training observations and then tested on unseen ones to properly evaluate its performance.

	Predicted 0	Predicted 1
True 0	40	0
True 1	0	40

Table 1: Confusion matrix with True labels (rows) and Predicted labels (columns).

The confusion matrix in Table 1 shows that all instances were correctly classified, with 40 true positives and 40 true negatives, and no misclassifications. This indicates a perfect separation between fake and real papers for this dataset.

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	40
1	1.00	1.00	1.00	40
Accuracy			1.00	80
Macro avg	1.00	1.00	1.00	80
Weighted avg	1.00	1.00	1.00	80

Table 2: Classification report of the model.

The classification report in Table 2 confirms this result: all evaluation metrics (precision, recall, and F1-score) achieved the maximum value of 1.00 for both classes. Additionally, the macro and weighted averages also reach

1.00, indicating that the performance is consistent across classes and unaffected by class imbalance.

The results show that all of the classifiers operate perfectly and can detect between authentic and fake documents. This could indicate that the distinction was so trivial that there was no need for a model to distinguish them.

Fake publications created by algorithms exhibit recurring patterns that make it easy to distinguish between real works written by humans and absurd content generated by machines. It should also be noted that SCIgen software was developed in 2005, and does not use evolved language models; instead, it applies context-free grammar, or random grammar rules.

This could have lead to a number of recurring patterns and incoherencies in the text, which are simple to recognize by modern machine learning models. This does not necessarily mean that more advanced text generators, like chat-GPT, will provide results with the same degree of accuracy.

It was also reported above that almost all publications produced with SCIgen have been deleted. Consequently, the range of fake articles in this study is limited. If all of them show the same trends, the models could have been severely effected.

The ROC curves for the four classifiers — SVM, Logistic Regression, Naïve Bayes, and Random Forest — are shown in Figure 1.

An Area Under the Curve (AUC) of 1.0 is achieved by all models, indicating complete capacity to distinguish between real and non authentic papers in the dataset. The ROC curves rise immediately towards the top-left corner, meaning that each classifier reaches a true positive rate of 1.0 while maintaining a false positive rate of 0.0 across all thresholds.

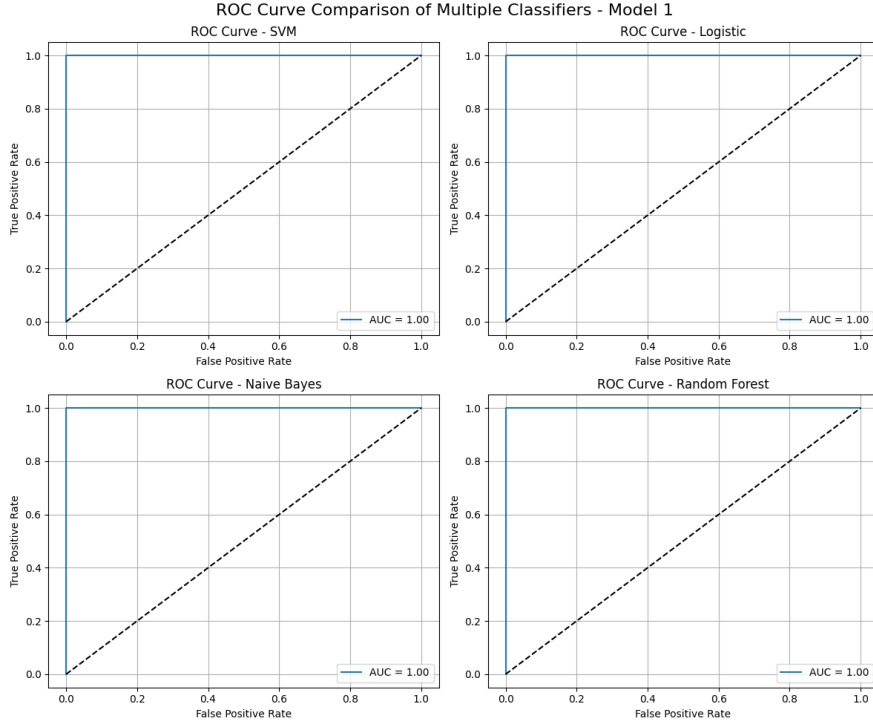


Figure 1: ROC curves comparison of the four classifiers - SVM, Logistic Regression, Naïve Bayes, and Random Forest.

2.5 N-grams: Pattern Recognition

The results made it evident that it might be too easy to distinguish the difference between authentic and generated papers. To advance the study further, an investigation was conducted to understand the patterns that differentiate these two categories. Based on repetition, it is possible to recognize a non real paper from an authentic one because “human style is essentially more ‘different’ and ‘rich’ than an artificial one” (Avros, Volkovich, 2018, p. 42). Fakes tend to repeat multiple internal structures and this redundancy is one of the reasons why models are able to classify them extremely well.

An n-gram analysis was necessary to determine the most common patterns in both type of documents. The analysis was extended to include 2-5 word combinations. In particular, with single terms (uni-grams), the model only captures isolated words (e.g., network, system), which often appear in both true and false cases, therefore not very discriminatory. On the other hand, 2-5 n-gram allows to capture the specific repetitive structure that show up in both papers.

The Table 3 shows the most influential n-grams.

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	vol pp	0.208126	proceeding workshop	-0.268824
2	ipv address	0.123427	pro ceeding	-0.246797
3	batch size	0.122926	proceeding conference	-0.228366
4	high performance	0.115872	consider harmful	-0.219000
5	ue antenna	0.104470	gigabit switch	-0.191527
6	quantum network	0.097916	context free	-0.187862
7	international conference	0.095163	cache coherence	-0.182612
8	real world	0.091634	technical report	-0.176043
9	end end	0.089868	context free grammar	-0.156652
10	control system	0.089447	free grammar	-0.156652
11	network traffic	0.089169	wide area	-0.156114
12	time slot	0.086644	proceeding fpca	-0.151975
13	ip address	0.079510	digital analog	-0.150075
14	large batch	0.079330	boolean logic	-0.148321
15	cross chain	0.077387	area network	-0.130890

Table 3: Comparison of top n-grams between REAL (left) and FAKE (right) papers.

Each paper in the dataset is converted into a TF-IDF vector, and each n-gram feature is given a weight — a numerical value that indicates how important a term is in the document.

Given that those weights were determined using SVM, the sign is important because it identifies the hyperplane that separates the two classes. Then, each document is placed on one side of the hyperplane, and based on that, the sign will be either positive or negative. In this case:

- **Positive weight** ($w > 0$): the presence of that n-gram pushes the classification towards **class 1**.
- **Negative weight** ($w < 0$): the presence of that n-gram pushes the classification towards **class 0**.

However, this does not mean that a positive n-gram can appear only on true papers; it just indicates that the model has learned that it occurs more frequently in those papers (or vice versa).

Furthermore, it is noticeable that n-gram values for fake papers are much higher than those for authentic papers. This confirms the idea that generated text (SCIgen, weak LLM, etc.) tend to use bizarre and repetitive collocations. The model is therefore more confident in classifying those as patterns of

fake papers than real ones, which tend to have lower values (most varied expressions).

Additionally, when analyzing the combinations, its noticeable that real papers tend to represent concrete technical terms and realistic contexts³. These expressions reflect real scientific concepts related to security, networking, physics, and machine learning. By contrast, fake papers display generic patterns that tend to recur frequently (pro ceeding, proceeding fpca, proceeding symposium, etc.).

Ultimately, the results maintain a 100% accuracy level, even if the analysis has been improved with 2-5 n-gram.

A SHAP summary plot (Figure 2) was created in order to confirm the function of n-grams in detecting authentic from fake publications.

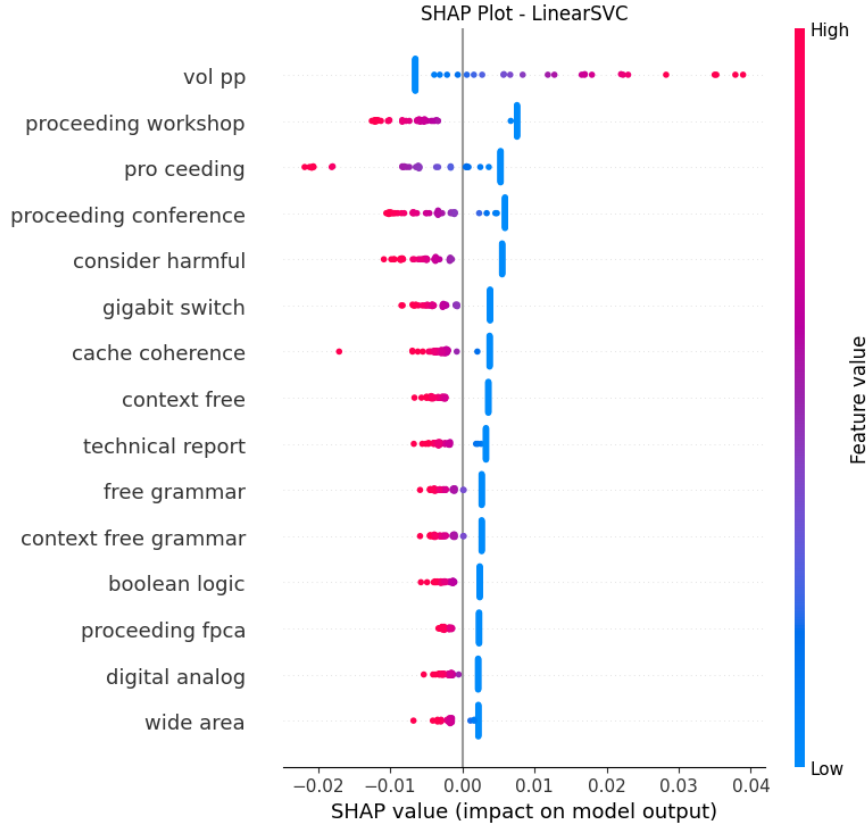


Figure 2: SHAP analysis

This demonstrates that the most significant features for the classification

³vol pp stands for volume and pages, frequently used in the bibliography.

model are, in fact, the same n-grams that are highlighted in Table 3. The consistency between SHAP-analysis and the SVM's statistical weights improves the conclusions' reliability.

3 Detecting advanced generated papers

In light of previous findings, this chapter’s goal is to evaluate the performance of classification models on more sophisticated generated papers. With the growing number of AI tools that can produce new types of text, the aim is to test the classifier’s robustness against more complex documents that more accurately reflect real-world situations.

3.1 Dataset

Since the dataset was thought to be unsuitable for drawing meaningful conclusions, it was considered to introduce more variety to fake papers. So, the non authentic documents will differ from the originals in several ways that will be described in the next section.

Two datasets are included in the study for a deeper analysis, and both have been pre-processed as describes in section 2.2.

It should also be considered that the data will be smaller than previously, due to issues that will be explained later.

3.1.1 Dataset Model 1

To improve the analysis, several modification are introduced in the first model. In particular the dataset is composed by:

- SCIGen generated papers modified with Chat-GPT.
- Documents realized through LLM.
- Real texts downloaded from arXiv.

Since the patterns produced by SCIGen might be too basic, given the time period in which it was created (2005), it seemed logical to introduce Large Language Models (LLM), which produce text that is much closer to human language. In order to produce sentences that closely mimic natural language, LLM are AI models trained on vast amounts of text data - such as books, web-pages, essays, and so forth - and learn the probabilities of word sequences.

A collection of 35 artificially generated academic papers was created using TinyLlama-1.1B-Chat⁴, a compact open-source language model l with 1.1

⁴Designed by the TinyLlama; project to be efficient and runnable on devices with limited computational and memory resources.

<https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0>

billion parameters.

After that, keywords from recent arXiv articles in the domains of systems and networks were analyzed to create customized prompts for various article sections. Using randomized text generation parameters (temperature = 0.9 and top-p = 0.92), the model produced additional content for standard academic parts (Introduction, Method, Results, etc.) while preserving consistency using a fixed random seed. To minimize redundancy, it was introduced variations in section sequencing, content length, writing styles and other elements.

Additionally, according to current trends, Chat-GPT is undoubtedly the most popular AI assistant powered by LLM; therefore, it was appropriate to add some paper generated from this interface. To do this, SCIdgen papers were utilized as a base to create with Chat-GPT since it was found in the prior chapter that they have a relatively simple structure to recognize. This method would improve the way articles are arranged, or at least, it introduce different patterns that might influence how well the model performs.

Using this procedure, only ten documents were obtained because the introduced changes were manually added, which combined with those created using TinyLlama-1.1B-Chat, resulted in a complete dataset of 45 fake papers.

To balance the set of data, a total of 45 true papers were downloaded from the arXiv website, using the same categories as above (cs.NI, cs.DC and cs.OS).

3.1.2 Dataset Model 2

For the second dataset, the overall structure is very similar but a different type of fake papers was incorporated. As a way to explore different insights, some authentic publications previously taken from the arXiv website were noised in a progressively way. Noise was consistently added to each real paper until all of the documents eventually looked like fakes. Through this incremental corruption process, the authentic papers were converted into fake ones, providing an alternative method for investigating classification insights. In all, 45 authentic texts were corrupted, resulting 90 generated articles (45 from LLM). Again, to maintain a balanced dataset, 90 new true papers were downloaded from arXiv.

3.2 Results

Before examining the outcomes, it is important to first explain how the models were trained. The dataset, as before, was split into two parts: 60% training data and 40% test data. Each model is trained using the training observations and then tested on unseen ones to properly evaluate its performance. In the first model the total observations are 90, of which 54 were used for training and the 36 for testing (equally split into 18 authentic and 18 fake). In the second one the dataset contains 180 observations, with 108 used for training and 72 for testing, again proportionally divided between authentic and fake papers.

Keep in mind that arXiv is a website that updates published articles on a regular basis, so even with a random seed set, the true papers you download and the outcomes you get could vary.

3.2.1 Results Model 1

In contrast to the trivial 100% results previously recorded, classification performance decreased when modified versions of SCIGen articles were substituted using chat-GPT and contemporary LLM-generated papers were introduced. This suggests that using advanced models to create fakes makes the task more difficult and realistic.

Specifically, while Random Forest still achieves perfect accuracy on the new dataset, simpler linear models such as Logistic Regression and SVM show reduced performance (83–89%), reflecting the closer resemblance of LLM text to authentic scientific writing. It’s interesting to note that Naïve Bayes is still very competitive (92%), suggesting that word-frequency patterns continue to provide strong discriminative cues.

These results highlight that detecting LLM-generated scientific papers is feasible but no longer trivial, and that future work should investigate larger datasets and more advanced embeddings to evaluate generalization more reliably.

However, it should be considered that even though the results are more promising, the size of it could be also a huge limitation. Since the test size was only 36 observations, the results are not conclusive (especially the 100% accuracy of RF). Furthermore, when using advanced text generators, the problem is not anymore trivial. There should be put more focus into the new advanced features introduced through those models.

False positives (authentic documents flagged as fake) are less concerning than false negatives (fake documents missed by the system), since our main

objective is detecting fraudulent content. It’s interesting to note that all models incorrectly classify the same documents, therefore accurately detecting the non-authentic ones continues to be the main objective.

	Pred. 0	Pred. 1
True 0	16	2
True 1	2	16

Table 4: Confusion matrix — SVM.

Class	Precision	Recall	F1-score	Support
0	0.89	0.89	0.89	18
1	0.89	0.89	0.89	18
Accuracy			0.89	36
Macro avg	0.89	0.89	0.89	36
Weighted avg	0.89	0.89	0.89	36

Table 5: Classification report for Support Vector Machine (SVM).

	Pred. 0	Pred. 1
True 0	14	4
True 1	2	16

Table 6: Confusion matrix — Logistic Regression.

Class	Precision	Recall	F1-score	Support
0	0.88	0.78	0.82	18
1	0.80	0.89	0.84	18
Accuracy			0.83	36
Macro avg	0.84	0.83	0.83	36
Weighted avg	0.84	0.83	0.83	36

Table 7: Classification report for Logistic Regression.

	Pred. 0	Pred. 1
True 0	17	1
True 1	2	16

Table 8: Confusion matrix — Naïve Bayes.

Class	Precision	Recall	F1-score	Support
0	0.89	0.94	0.92	18
1	0.94	0.89	0.92	18
Accuracy			0.92	36
Macro avg	0.92	0.92	0.92	36
Weighted avg	0.92	0.92	0.92	36

Table 9: Classification report for Naïve Bayes.

	Pred. 0	Pred. 1
True 0	18	0
True 1	0	18

Table 10: Confusion matrix — Random Forest.

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	18
1	1.00	1.00	1.00	18
Accuracy			1.00	36
Macro avg	1.00	1.00	1.00	36
Weighted avg	1.00	1.00	1.00	36

Table 11: Classification report for Random Forest.

The results are shown also through graphical representation of the ROC curve, Figure 3. A lower AUC value reflect a higher challenge in classifying, even though only by a small amount. Nevertheless, all the models perform greatly, Random Forest achieving perfect separation ($AUC = 1.00$), which suggests that despite their increased linguistic sophistication, LLM-generated papers still exhibit detectable lexical and structural patterns.

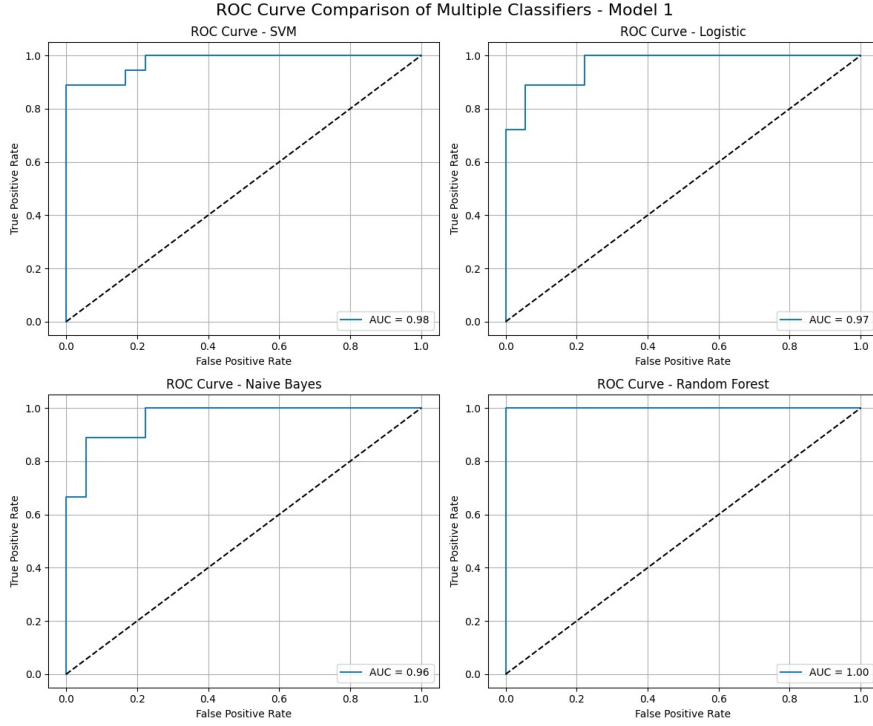


Figure 3: ROC curves comparison of the four classifiers (SVM, Logistic Regression, Naïve Bayes, and Random Forest).

3.2.2 Results Model 2

Compared to the dataset in model 1, the one in the second model is less trivial because the fake papers are derived also from corrupted real documents, making them more similar to authentic texts. Despite this, they still exceed 90% accuracy.

The robustness of Random Forest suggests that tree models are able to exploit subtle patterns even when the boundary between classes is blurry. Naïve Bayes suffers more from the noisy environment ($AUC = 0.94$), indicating that its independence assumption is less suitable for corrupted text.

Overall, the classifiers' outstanding performance shows that distinguishable statistical signals exist even with increasingly complex fake generating techniques. Also here, the misclassified documents are the same across all the different methods.

The ROC curve representation further supports Random Forest's robustness, demonstrating that it outperforms linear or probabilistic models in article classification.

	Pred. 0	Pred. 1
True 0	33	3
True 1	2	34

Table 12: Confusion matrix — SVM (L2).

Class	Precision	Recall	F1-score	Support
0	0.94	0.92	0.93	36
1	0.92	0.94	0.93	36
Accuracy			0.93	72
Macro avg	0.93	0.93	0.93	72
Weighted avg	0.93	0.93	0.93	72

Table 13: Classification report — SVM (L2).

	Pred. 0	Pred. 1
True 0	31	5
True 1	1	35

Table 14: Confusion matrix — Logistic Regression.

Class	Precision	Recall	F1-score	Support
0	0.97	0.86	0.91	36
1	0.88	0.97	0.92	36
Accuracy			0.92	72
Macro avg	0.92	0.92	0.92	72
Weighted avg	0.92	0.92	0.92	72

Table 15: Classification report — Logistic Regression.

	Pred. 0	Pred. 1
True 0	33	3
True 1	5	31

Table 16: Confusion matrix — Naïve Bayes.

Class	Precision	Recall	F1-score	Support
0	0.87	0.92	0.89	36
1	0.91	0.86	0.89	36
Accuracy			0.89	72
Macro avg	0.89	0.89	0.89	72
Weighted avg	0.89	0.89	0.89	72

Table 17: Classification report — Naïve Bayes.

	Pred. 0	Pred. 1
True 0	36	0
True 1	0	36

Table 18: Confusion matrix — Random Forest.

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	36
1	1.00	1.00	1.00	36
Accuracy			1.00	72
Macro avg	1.00	1.00	1.00	72
Weighted avg	1.00	1.00	1.00	72

Table 19: Classification report — Random Forest.

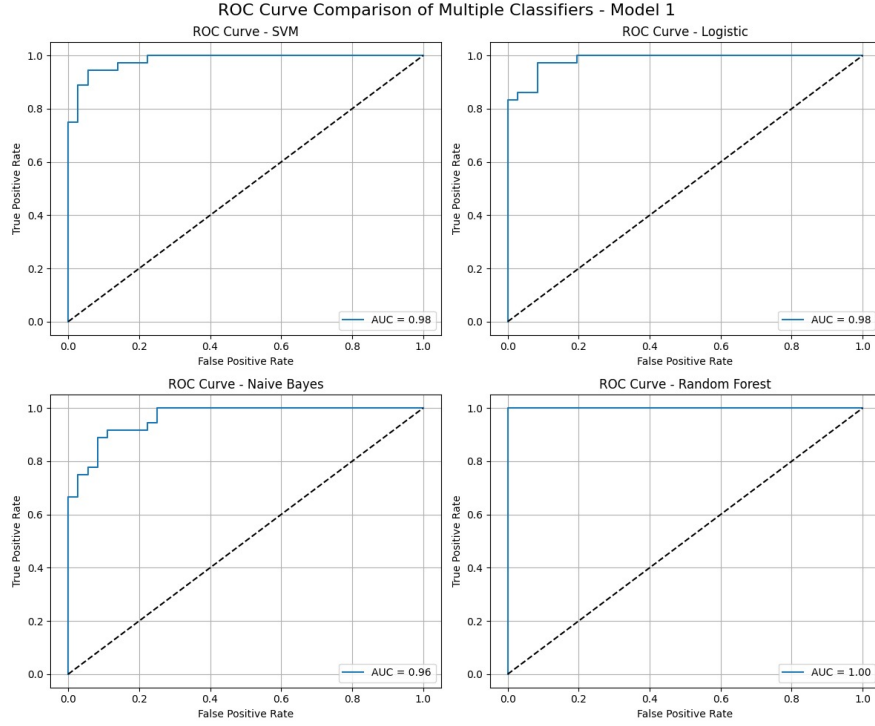


Figure 4: ROC curves comparison of the four classifiers (SVM, Logistic Regression, Naïve Bayes, and Random Forest).

3.3 N-grams: Pattern Recognition

In order to enhance the study, both models were subjected to the n-grams method, which aims to find recurrent patterns in the different texts that would allow better classification. Combinations of two to five consecutive words are taken into account in the research, which highlights text structural tendencies. SHAP summary plots (Figure5 and Figure6) were created in order to confirm the function of n-grams in detecting authentic from fake publications.

The Table20 shows the most influential n-grams for the Model 1.

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	et al	0.467073	mixed precision	-0.366800
2	vol pp	0.447904	holo artisan	-0.342372
3	batch size	0.353157	threat validity	-0.284088
4	online available	0.316916	system perspective	-0.278062
5	bad case	0.286687	reference reference	-0.242812
			omit brevity	
6	smart contract	0.286330	distribute system	-0.242812
			september abstract	
7	step size	0.284732	synthetic author	-0.242812
			institute distribute	
			system	
8	formal specification	0.245368	synthetic document	-0.242812
9	large batch	0.242016	institute distribute	-0.242812
			system	
10	top top	0.230980	institute distribute	-0.242812
			system september	
11	international conference	0.221889	system september	-0.242812
12	global model	0.216582	institute distribute	-0.242812
			system september	
			abstract	
13	arxiv preprint	0.206887	author institute	-0.242812
			distribute system	
			september	
14	preprint arxiv	0.206887	reference reference	-0.242812
15	arxiv preprint arxiv	0.206887	reference reference	-0.242812
			omit	

Table 20: Comparison of top n-grams between REAL (left) and FAKE (right) papers.

Because the dataset is different from the prior study (section 2.5), the word sequences are also different. In spite of this, the patterns of fake papers — such as reference reference, synthetic author, or synthetic perspective — remain less regular or repetitive than those of real publications.

While, Table21 shows the most influential n-grams for the Model 2.

Rank	N-gram (Real)	Weight	N-gram (Fake)	Weight
1	vol pp	0.514356	doe roe scalable	-0.762000
2	et al	0.311016	doe roe	-0.762000
3	international confer- ence	0.274895	roe scalable	-0.762000
4	leaf switch	0.253222	large scale system	-0.762000
5	proceeding th	0.248256	scale system	-0.724525
6	porto alegre	0.246575	threat validity	-0.348859
7	online available	0.229068	holo artisan	-0.346426
8	share memory	0.220238	experimental setup	-0.345480
9	ieee transaction	0.186775	system perspective	-0.344376
10	rio grande	0.184166	reference reference	-0.330951
11	preprint arxiv	0.183555	large scale	-0.314842
12	rio grande sul	0.181577	system perspective synthetic author	-0.311172
13	grande sul	0.181577	omit brevity syn- thetic	-0.311172
14	two way	0.179981	system perspective synthetic	-0.311172
15	arxiv preprint	0.176071	perspective syn- thetic author	-0.311172

Table 21: Comparison of top n-grams between REAL (left) and FAKE (right) papers.

Here, the patterns of non-authentic texts tend to be slightly less inconsistent, as they contain noisy true data documents. Groups of words like "doe roe," however, are evident and provide unmistakable proof that the article has been generated. The phrases "John Doe" and "Jane Roe" are examples of fake author names that are used in computer manufactured publications to imitate professional references.

The results, in both models, are that with n-grams the classifier works perfectly.

	Predicted 0	Predicted 1
True 0	18	0
True 1	0	18

Table 22: Confusion matrix with True labels (rows) and Predicted labels (columns) - Model 1.

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	18
1	1.00	1.00	1.00	18
Accuracy			1.00	36
Macro avg	1.00	1.00	1.00	36
Weighted avg	1.00	1.00	1.00	36

Table 23: Classification report - Model 1.

	Predicted 0	Predicted 1
True 0	35	1
True 1	0	36

Table 24: Confusion matrix with True labels (rows) and Predicted labels (columns) - Model 2.

Class	Precision	Recall	F1-score	Support
0	1.00	0.97	0.99	36
1	0.97	1.00	0.99	36
Accuracy			0.99	72
Macro avg	0.99	0.99	0.99	72
Weighted avg	0.99	0.99	0.99	72

Table 25: Classification report - Model 2.

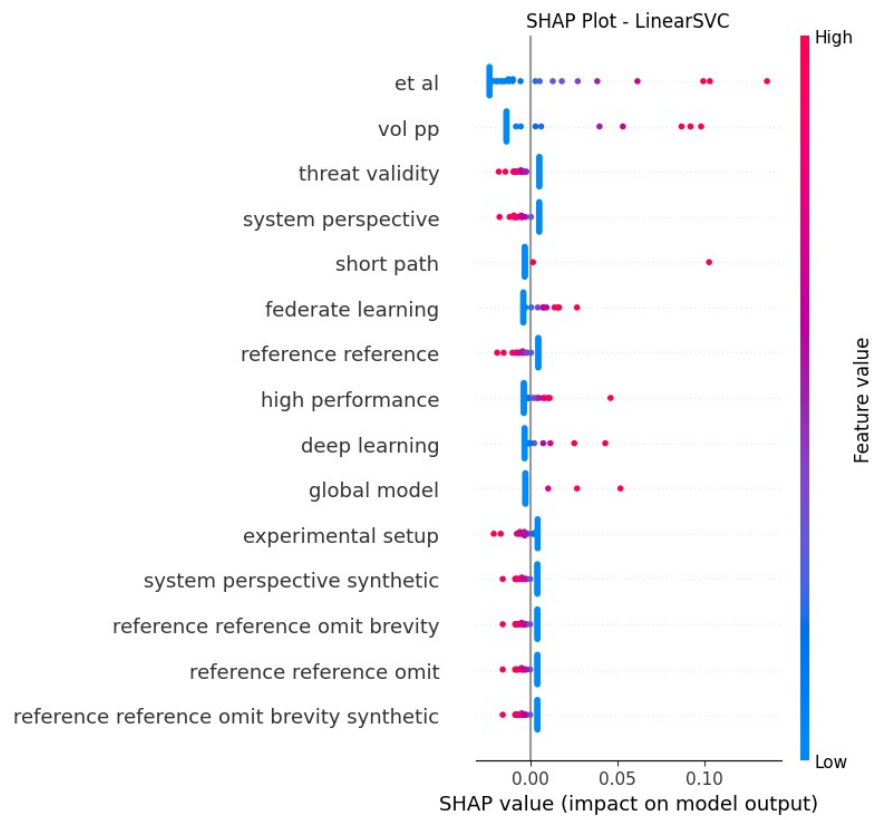


Figure 5: SHAP analysis Model 1

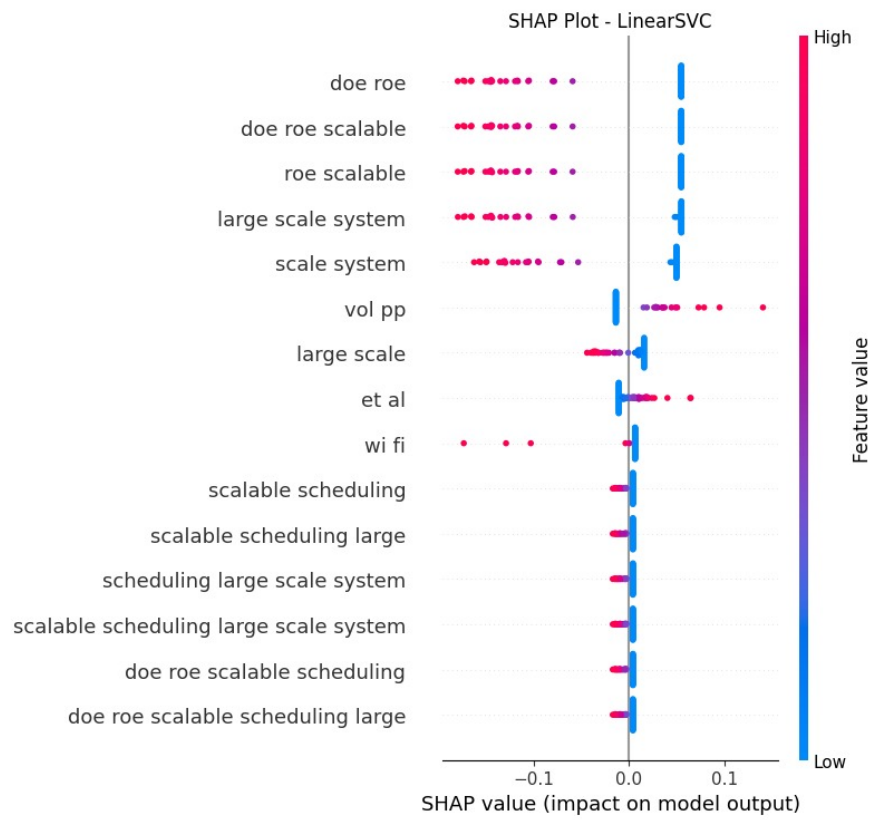


Figure 6: SHAP analysis Model 2

4 Conclusions

It is evident from the results that the SCIgen approach alone makes it easier to identify between documents, but the introduction of noise through new developments complicate the analysis. Although still rather good, the accuracy can actually drop from 100% across all categorization models - SVM, Logistic Regression, Naïve Bayes, and Random Forest - to between 83% and 93%. This suggests that methods like LLM can "trick" algorithms like SVM and Logistic Regression by producing texts that are more similar to natural language. Random forest, instead, is still robust in the results.

In order to advance the analysis in the project's first study - Detecting SCIgen Generated Papers - n-grams were included. This confirmed that, in contrast to real texts, which have more cohesive expressions, fake texts typically contain "nonsensical" word combinations (e.g., proceeding workshop, pro ceeding, proceeding symposium).

Using n-grams makes it possible to comprehend the model's decision-making process when classifying a document as true or false.

A first limitation, that should be mentioned again, is that the study uses a smaller dataset because only 100 of the papers produced by SCIgen - all of which were "written" by Ike Antkare, a fake author — have been utilized, and the majority of the publications have already been deleted from the most popular platforms (like Google Scholar).

A second restriction of this project is that some fake documents were generated using TinyLlama-1.1B, a small open-source model that runs on CPU without requiring GPU hardware. While this choice ensures broad compatibility and zero costs, it comes with trade-offs: slower generation times due to extensive text requirements and reduced quality compared to premium models, including limited coherence and imprecise length control.

Additionally, the SCIgen papers enhanced via Chat-GPT represented just a limited sample of 10 papers, as they needed to be manually processed one at a time.

Note that the study was limited to only TF-IDF for classification models. Several advanced feature engineering approaches could improve classification beyond the basic TF-IDF such as: part-of-speech tag distributions which could capture grammatical patterns, word embedding-based similarity measures between sections, BERT embeddings could provide richer contextual representations than TF-IDF. These enhanced features could potentially improve detection accuracy by capturing deeper linguistic characteristics of

both authentic and non authentic writing.

It is important to keep in mind that publications on arXiv are updated daily, so the results of the analysis may differ slightly when the code is sent.

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